A market-making agent for prediction markets, benchmarked against the published optimum

Practice of QCF | Fall 2026 | Non-industry project proposal

Team members

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Problem statement

We will study whether reinforcement learning (RL) can improve prediction-market quoting when a theoretical strategy uses incorrect assumptions. Binary contracts settle at $1 or $0. A market maker posts buy and sell prices to earn the spread, but faces inventory losses and adverse selection when better-informed traders trade against those quotes. Our question is how much an optimal control strategy deteriorates under informed trading and sudden information events, and whether an RL agent can reduce that deterioration. The core project covers one binary contract at a time. Its contribution is a controlled comparison of quoting policies using empirically informed simulations.

Related literature

- **Optimal Market Making in Prediction Markets.** Feil and Nendel (2026) derive and verify an optimal quoting policy using a Hamilton-Jacobi-Bellman (HJB) equation with inventory and settlement-risk penalties. We will reproduce its numerical solution; optimality holds within its stated model.
- **Adverse Selection in Prediction Markets.** Bartlett and O'Hara (2026) document informed price impact using 41.6 million Kalshi trades and differences across market types. Their estimates guide our adverse-selection calibration.
- **Volatility in Prediction Markets.** Xi et al. (2026) connect volatility to probability, time to resolution and order flow. Their findings guide stress scenarios with discrete information arrivals.
- **ParlayMarket.** Rana et al. (2026) study automated market making for joint contracts. This provides related work for a future extension; correlated contracts remain outside our semester scope.

Methodology

Baseline reproduction. Implement the price process, quote-dependent executions, inventory limits and binary settlement from Feil and Nendel, then solve the HJB numerically. Check numerical convergence and reproduce selected quoting and risk results before training RL.

Controlled extensions. Add traders with private signals linked to subsequent price changes or settlement. Match simulated signed-flow price-impact estimates to published Kyle's lambda targets using the same five-minute aggregation and units. Test sensitivity ranges because lambda does not uniquely identify informed arrival rates. Separately introduce probability-consistent information jumps. Evaluate four environments: the original model, informed trading only, jumps only, and both.

Learning policy. Train a Proximal Policy Optimization agent using public price, inventory, time remaining, recent volatility and signed order flow. Actions select bid and ask quotes. Reward is the change in marked portfolio value, less running inventory penalties, with a final settlement adjustment and terminal inventory penalty. It will reproduce the HJB objective in the original environment.

Evaluation. Compare RL with the original HJB, a recalibrated HJB and a simple inventory-and-flow quoting rule. Apply common information availability, inventory limits and evaluation objectives. Introduce fees, tick sizes and execution delays consistently in stress tests. Split calibration, validation and test data chronologically by event, keep related contracts together, and evaluate on unseen scenarios across at least five training seeds. Report objective value, net profit, worst-5% average losses, drawdown and inventory exposure, with confidence intervals clustered by event.

Data needed and feasibility

Kalshi is the primary source. We need trade timestamps, prices, sizes and taker direction; minute bid/ask candles; market categories and identifiers; closing times, resolution rules and settlement outcomes; and dated fee and tick-size information. We will target 50–100 standalone contracts across economics and sports over a recent three-month window, subject to coverage. Category-specific calibration will avoid treating unlike markets as interchangeable.

We tested data availability by sending requests to the public Kalshi and Polymarket APIs. Kalshi returned an order book with 10 levels on each side, 100 trades with direction information and 11 one-minute price candles. Polymarket returned a two-sided order book, 100 trades and 1,441 price-history observations. The test confirms that sample data can be downloaded without authentication; complete historical coverage still needs verification. We will use Kalshi's public trades API (/markets/trades) and candlestick API, with historical endpoints for archived records. The collector will retrieve all pages of results and align timestamps, price units and trade direction. Polymarket remains an optional external comparison.

Expected outcomes

Deliverables are a documented data pipeline, a verified HJB implementation, an informed-trading and jump simulator, an RL quoting policy, and a reproducible performance comparison. We will identify whether and when adaptation improves the profit-risk trade-off and where the theoretical strategy remains competitive. RL superiority is not assumed. Biweekly reports will document progress and individual contributions; the final report and presentation will summarize findings and limitations.

Expected challenges

- **Execution uncertainty.** Historical candles and snapshots do not reveal queue position or whether our hypothetical orders would have filled. We will use conservative execution assumptions and sensitivity tests, limiting conclusions to simulated performance.
- **Data quality and coverage.** Sparse histories, changing API fields and quotes recorded at different times require validation before calibration.
- **Model calibration.** Price impact does not uniquely identify private information, and jumps must preserve valid probabilities and coherent settlement.
- **Generalization.** RL may overfit simulator assumptions. Unseen test scenarios and simple adaptive baselines will help assess this risk.
- **Time and computing limits.** If history or compute is insufficient, we will reduce the contract sample and prioritize HJB replication plus informed trading, retaining jumps as the next extension.